
Auditing a single-cell reporting rule: coverage, false positives and target choice

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Abstract

1 A reporting rule should be evaluated against the quantity and uncertainty it claims
2 to report. We audit a 50-cell rule in two colorectal single-cell cohorts that had been
3 justified by point-estimate recovery. In 5,000 simulations per setting, its nominal
4 95% percentile intervals reject a true null in 9.8–32.2% of studies at the reporting
5 boundary. A paired comparison with Welch intervals improves effect coverage but
6 leaves null rejection at 8.9–30.8%; changing the interval alone does not repair the
7 rule. We also show how a sampled reference inflates apparent coverage, explain
8 why sparse samples can produce misleadingly high detection, and document grid-
9 dependent threshold selection. A synthetic external-control example illustrates a
10 related mistake: a reference for the wrong population favors an estimator of the
11 wrong effect. The practical lesson is to specify the target, test null and alternative
12 performance together, and evaluate an interval and its reporting threshold as a
13 combined rule.

14 1 Motivation: recovery and decision-rule validation

15 A cell-count threshold is useful only if the estimates it permits have adequate uncertainty. We
16 audit a 50-cell reporting rule in colorectal single-cell analysis, testing coverage, false positives and
17 threshold stability. We then change the interval method while holding the simulated samples fixed,
18 to distinguish a failure of this implementation from a problem that a simple interval substitution
19 resolves. The contribution is an empirical case study of a previously overstated validation claim in an
20 unpublished project, with reproducible comparisons of targets and interval methods.

21 Our motivating application separates two explanations for differentiation-marker loss in colorectal
22 cancer: fewer marker-producing cells (a *compositional* change), or lower output from the cells that
23 remain (an *intrinsic* change). These explanations suggest different biological interpretations. An
24 empirical comparison of per-cell means requires the relevant population to be observed in both tissues.
25 Let f_N, f_T denote the relevant cell fractions in reference and diseased tissue, and m_N, m_T their
26 mean per-cell output. A reference-based decomposition, related to demographic standardisation and
27 Oaxaca–Blinder decompositions [Kitagawa, 1955, Oaxaca, 1973, Blinder, 1973], is

$$\underbrace{(f_T - f_N)m_N}_{\text{compositional}} + \underbrace{f_N(m_T - m_N)}_{\text{intrinsic}} + \underbrace{(f_T - f_N)(m_T - m_N)}_{\text{interaction}} = f_T m_T - f_N m_N. \quad (1)$$

28 The interaction is reported separately. If no relevant cells are observed in an arm, its sample mean is
29 undefined; the identity cannot supply the missing mean. In a synthetic population with no such cells,
30 the corresponding conditional mean itself has no referent. Substituting zero in either case turns an
31 unavailable contrast into an apparent effect.

32 The implemented reporting rule returns an estimate at $n \geq 50$, flags a wide interval at $20 \leq n <$
33 50, and returns NOT ESTIMABLE below 20, where n is the relevant-cell count in diseased tissue.

Unavailable reported contrasts are stored as None, not zero. The diagnostic oracle calculations examined below also retain raw arithmetic before this reporting gate. We evaluate the boundary directly and retain pre-gate intervals at smaller counts to diagnose the curve’s behavior.

Relation to existing validation methods. Simulation-study methodology already separates estimands, methods and performance measures [Burton et al., 2006, Morris et al., 2019]. Generator–analysis alignment is discussed in congeniality [Meng, 1994], plasmode and semi-synthetic benchmarks [Franklin et al., 2014, Curth et al., 2021], interventional evaluation [Gentzel et al., 2019] and the inverse crime [Wirgin, 2004, Kaipio and Somersalo, 2007]; generators can also disadvantage good estimators [Shaw et al., 2026]. We apply these distinctions to the recorded reporting-rule decision. For clinical trials, external-control validation already examines bias and type-I error using held-out studies [Ventz et al., 2019]. Prognostic adjustment with historical data retains trial randomisation and evaluates precision gains [Schuler et al., 2022]; target-trial emulation states the causal question and design explicitly [Hernán and Robins, 2016]. These methods motivate the clinical-simulation validation questions in Section 3. The empirical study uses single-cell pools; a separate synthetic external-control experiment illustrates target-population mismatch. Neither validates a patient world model. Simulation-based calibration evaluates posterior ranks [Talts et al., 2018, Modrák et al., 2025]. Appendix E provides further context.

2 Calibration study and results

The claim being re-evaluated. An internal assessment (Artifact A) treated the 50-cell boundary as validated because recovery ratios were 0.85–1.08 above it and 1.18 in the 20–49-cell band, using six draws per setting at 1,200 cells per arm. The design record (Artifact B) instead required at least 90% interval coverage and 80% exclusion of zero at a halving of expression. Appendix E and the anonymized supporting record make this mismatch inspectable.

2.1 Simulation and inferential targets

The design document fixed a detectable halving of per-cell output ($s = 0.5$), coverage of the target in at least 90% of defined outputs, and discrimination of at least 80%, defined as the probability that the interval excludes zero at that effect. The implementation scored coverage against a drawn-reference benchmark; we retain that score and add coverage against the generating pool mean. The candidate *ok* is the first evaluated count or bin meeting both criteria; *wide* is the first meeting coverage alone. The committed values are 50 and 20. The first-crossing definition does not require larger counts to qualify.

The calibration uses paired subsets of the published colorectal atlas [Lee et al., 2020]: ten patients from GSE132465 (SMC) and six from GSE144735 (KUL3). We analyse GUCA2A in the differentiated population under fixed lineage labelling. Each draw samples, with replacement, 2,000 cells per arm from an eligible two-patient pool; the reference mature fraction is $f_N = 0.40$. The implemented pool combines cells from both tissues before drawing either synthetic arm. Because the design document did not specify tissue eligibility, we also evaluate a reference-tissue-only pool. Both arms sample the same eligible pool, with an imposed expression shift. A nominal 95% percentile interval resamples mature cells in both arms, with fractions held fixed and 200 inner draws unless stated otherwise. Undefined intervals are excluded from rate denominators and reported as abstentions (Appendix B).

The stored “parametric” intrinsic reference is

$$i_{\text{req}}(D_N; s) = f_N(s - 1)\bar{Y}_N, \quad (2)$$

where $\bar{Y}_N$ is the mature-cell mean of the *drawn* reference arm. Let P be the eligible two-patient source pool, M_P its mature-cell count, and Y_j its unshifted target-gene counts. The population parameter of this finite empirical distribution is instead

$$\mu_P = \frac{1}{M_P} \sum_{j \in P, \text{mature}} Y_j, \quad \theta_P(s) = f_N(s - 1)\mu_P. \quad (3)$$

We compute μ_P from all eligible source cells before either synthetic arm is sampled. Given P , the generator has $E[\bar{Y}_N] = \mu_P$ and $E[\bar{Y}_T] = s\mu_P$, making Equation 3 its true intrinsic mean contrast. It is fixed across cell draws from that pool. Patient-pair selection changes P ; reported coverage

averages this conditional-pool question over the selected pairs. An all-patient pooled mean would use a different population from the generator’s two-patient draw. Neither target supplies inference over new patients.

Two complementary runs separate historical selection from direct evaluation. The controlled eight-seed grid retains inclusion of Equation 2. An initial 200-replicate diagnostic reproduced its first-seed results and added fixed-pool coverage and null rejection. The precision follow-up uses 5,000 fresh studies per cohort, pool, count (5, 50, 100, 800) and shift ($s = 0.5, 1$), split across five seed streams: 160,000 studies in total. It samples the exact marginal law of mature-cell counts in the original generator; non-mature cells do not enter this contrast. On each study we compare the original 200-resample percentile interval with a Welch–Satterthwaite interval [NIST/SEMATECH]. This standard unequal-variance t approximation is a comparator, not an assumed solution for sparse counts. All 320,000 intervals are finite. The largest binomial Monte Carlo standard error is 0.71 percentage points, conditional on the empirical source pools. Appendix A specifies the sampling and interval calculations.

2.2 Fixed-pool coverage and false positives

The larger run resolves the 50-cell coverage comparison (Table 1). Percentile coverage is below the recorded 90% criterion in three conditions, whose Wilson upper endpoints are below 90%. SMC/reference has 91.0% coverage, with its lower endpoint above 90%. Thus the rule does not fail coverage uniformly. It does fail the combined assessment: null rejection is 9.8–32.2%, above the nominal 5% in every condition, and detection is 47.5–76.2%, below the required 80%. All four detection upper endpoints are below 80%; all four null-rejection lower endpoints exceed 5%. These intervals measure simulation precision, not uncertainty over additional patients.

Table 1: At the 50-cell reporting boundary, changing intervals improves coverage but does not restore the required performance. Rates (%) and 95% Wilson Monte Carlo intervals use 5,000 studies per cohort, pool and effect. Methods share the same samples. Coverage targets the fixed-pool effect at $s = 0.5$; null rejection uses $s = 1$; detection uses $s = 0.5$. Targets are at least 90% coverage, nominal 5% null rejection and at least 80% detection.

Cohort / pool	Interval	Coverage	Null rejection	Detection
SMC / pooled	Percentile	87.1 [86.2, 88.0]	25.4 [24.2, 26.6]	51.1 [49.7, 52.4]
SMC / pooled	Welch	90.2 [89.3, 91.0]	23.5 [22.3, 24.7]	50.7 [49.3, 52.1]
SMC / reference	Percentile	91.0 [90.2, 91.8]	9.8 [9.0, 10.7]	76.2 [75.0, 77.3]
SMC / reference	Welch	92.5 [91.8, 93.2]	8.9 [8.1, 9.7]	77.6 [76.5, 78.8]
KUL3 / pooled	Percentile	84.7 [83.6, 85.6]	32.2 [31.0, 33.5]	49.0 [47.6, 50.4]
KUL3 / pooled	Welch	89.7 [88.8, 90.5]	30.8 [29.5, 32.1]	49.0 [47.6, 50.3]
KUL3 / reference	Percentile	86.3 [85.3, 87.2]	19.0 [18.0, 20.1]	47.5 [46.1, 48.8]
KUL3 / reference	Welch	88.1 [87.2, 89.0]	18.8 [17.7, 19.9]	48.2 [46.8, 49.6]

At 800 cells, evaluating the percentile intervals against the sampled reference gives 98.9–99.6% inclusion, compared with 92.3–93.6% coverage of the fixed-pool effect. The paired differences are 5.8–6.8 percentage points with Monte Carlo standard errors of 0.33–0.36 points. Only the scoring target changes. At five cells the direction reverses; shared randomness need not inflate inclusion at every sample size.

The increasing discrepancy at large tumour-arm counts has an analytic explanation. The estimate is $\hat{\theta} = f_N(\bar{Y}_T - \bar{Y}_N)$, whereas its error against the drawn benchmark is $\hat{\theta} - i_{\text{req}} = f_N(\bar{Y}_T - s\bar{Y}_N)$. For independent arm draws conditional on P , their error variances differ by

$$\text{Var}(\hat{\theta} - \theta_P \mid P) - \text{Var}(\hat{\theta} - i_{\text{req}} \mid P) = f_N^2(1 - s^2) \frac{\sigma_P^2}{n_N}, \quad (4)$$

where σ_P^2 is the source-pool expression variance and $n_N = 800$. At $s = 0.5$, the drawn benchmark removes three quarters of the reference-arm variance contribution from the error being assessed, while the bootstrap continues resampling both arms. This explains why near-perfect benchmark inclusion can accompany lower coverage of the generating parameter. Skewed finite-sample errors still determine the actual percentile-interval rates.

116 The new runs assess four specified counts, rather than selecting a replacement threshold. The
 117 eight-seed grid below remains scored against its original benchmark and cannot be reinterpreted as
 118 fixed-pool calibration.

119 2.3 Changing the interval helps, but does not repair the rule

120 Welch intervals increase fixed-pool coverage at 50 cells by 1.5–5.0 percentage points on the same
 121 draws (paired Monte Carlo standard errors 0.23–0.32 points). They also reduce null rejection, but
 122 it remains 8.9–30.8% and every lower confidence endpoint exceeds 5%. Detection remains below
 123 80% even at the upper confidence endpoints. Wider intervals improve some operating characteristics
 124 without satisfying the joint requirements. This rules out this particular substitution as a repair at 50
 125 cells; it does not rule out other methods or validate a higher threshold. Null and alternative evaluation
 126 must accompany any proposed repair.

127 **Why detection can fall as cell count rises.** The apparent detection curve combines effect sensitivity
 128 with a changing false-positive rate. Figure 1 reproduces the pooled-draw trough with greater precision.
 129 For KUL3, percentile detection falls from 49.0% at 50 cells to 44.0% at 100, then rises to 57.7% at
 130 800; null rejection falls from 32.2% to 20.3% to 6.5%. At 50 and 100 cells, the median sample-based
 131 standard error is respectively 0.66 and 0.86 times the exact pool-conditional standard deviation. The
 132 fraction of draws with that ratio below one half falls from 29.1% to 7.6%. These diagnostics support
 133 a loss of spuriously confident rejections as the sampled variance becomes better represented, before
 134 increasing signal-to-noise dominates. They do not attribute every change to one mechanism. In
 135 particular, all-zero tumour samples occur in only 0.32% of the 50-cell draws and none at 100: the
 136 trough is not explained by all-zero samples alone.

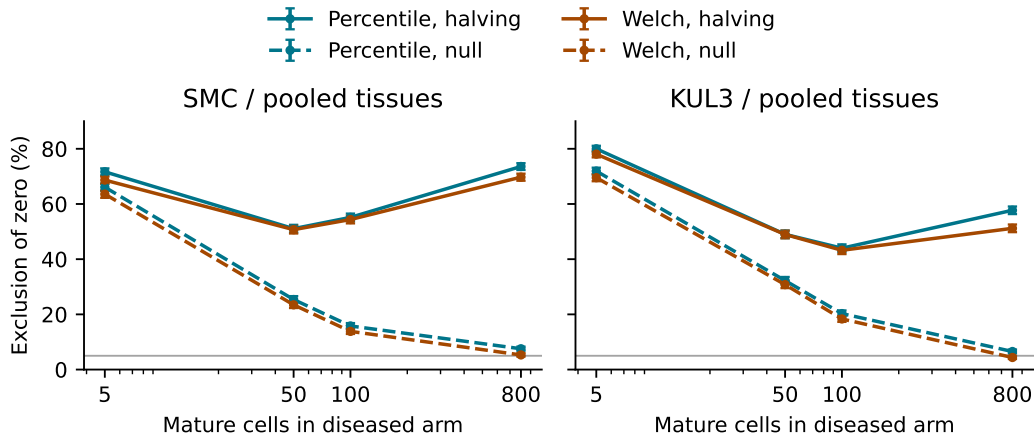


Figure 1: Detection must be read alongside null rejection. Pooled-tissue simulations use 5,000 studies per point; bars are 95% Wilson Monte Carlo intervals. Solid curves use a halving of expression and dashed curves the null. The horizontal line marks nominal 5% rejection. Five-cell intervals are diagnostic calculations before the reporting gate, which abstains below 20. Both methods use the same samples; their apparent sensitivity at sparse counts accompanies excessive false positives.

137 The five-cell diagnostic makes the problem especially clear: pooled percentile null rejection is 66.1%
 138 in SMC and 71.9% in KUL3. Under the halving, 46.2% and 41.9% of tumour samples are all zero.
 139 Resampling cannot recover an unobserved positive tail, although the reference arm still contributes
 140 uncertainty. The original reporting gate already abstains here; the 50-cell failures concern returned
 141 answers directly.

142 2.4 Count grids change candidate cutpoints

143 Historical grids returned reference-only binned candidates of 100, 90 and 70, but changed random
 144 streams, bin boundaries and replicate budgets together (Appendix B). To isolate these choices, we
 145 simulated the union of counts once per cohort, pool and each of eight seeds, with 200 outer replicates
 146 and 200 inner bootstrap draws. Every grid summary reuses the same outputs at shared counts. The

147 committed grid reaches $\{0, 20, 40, 100, 200, 400, 800\}$; filled and dense grids add count support.
148 Fixed bins share boundaries across grids; adaptive bins recalculate them. Binned candidates are
149 median counts within the selected bin, not estimated crossing locations.

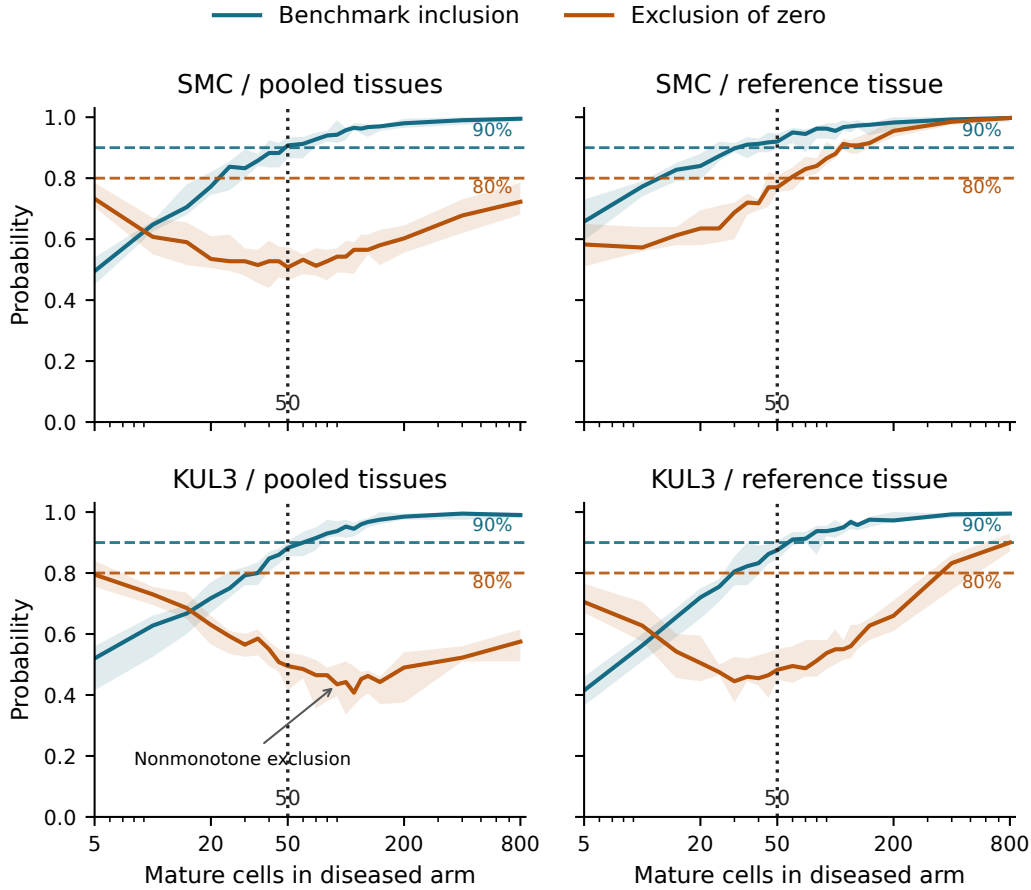


Figure 2: Direct evaluation on the controlled union of counts, without binning. Rows show cohorts; columns show draw pools. Curves are median rates over eight seeds and bands span their minima and maxima, conditional on the observed cohorts. Dashed lines mark the 90% benchmark-inclusion and 80% discrimination targets; the vertical line marks the committed count of 50. Rates condition on defined intervals and coverage uses the draw-dependent reference in Equation 2. Zero-cell settings have no defined interval and are omitted. No pooled-cell candidate qualifies. Per-count reference-only candidates vary by seed and cohort (Table 2).

150 With common fixed bins, SMC returns 100 on the committed grid and 70 on both expanded grids in
151 all eight seeds. Adaptive bins return 65 on seven filled-grid seeds and 90 on one, while the dense
152 grid returns 70 throughout. KUL3 instead returns 400 on six seeds and 800 on two, with all grid and
153 binning schemes agreeing within each seed. Pooled draws yield no joint candidate in either cohort.
154 The largest binomial Monte Carlo standard error in this grid is 0.035.

Table 2: Joint coverage-and-discrimination candidates from the controlled union-grid run. Entries give candidates with seed frequencies in parentheses where they vary; otherwise all eight seeds agree. Every grid summary reuses the same draws at shared counts. “None” means no tested count or bin qualifies.

cohort / pool	grid	fixed bins	adaptive bins	per count
SMC / reference	committed	100	100	100
	filled	70	65 (7), 90 (1)	50–80
	dense	70	70	45–80
KUL3 / reference	all three	400 (6), 800 (2)	same	same
Both / pooled	all three	None	None	None

At the committed count of 50, SMC reference-only discrimination ranges from 75.5% to 81.0% across seeds, with both criteria met on two of eight seeds. KUL3 reference-only discrimination is only 45.0–55.0% at that count. The reporting boundary therefore misses the discrimination requirement in six of eight SMC seeds and every KUL3 seed.

The per-count summaries in Table 2 supply a useful check on the bin medians. In SMC, the filled grid’s first qualifying actual count is 50 on two seeds, 60 on five and 80 on one; the dense grid adds a candidate of 45 on one seed. Thus a stable binned candidate of 70 is not evidence that 70 is a uniquely determined biological threshold. A bin pools outcomes over several count settings, whereas a per-count rate addresses only that setting. Fixed bins hold the boundaries constant, but adding settings can still change both a bin’s median count and its aggregate performance.

2.5 First qualification can reverse at larger counts

Three of eight SMC/reference seeds in the complete saved union grid fail a criterion after first qualifying. This matters for a rule that reports all counts above a scalar boundary: passing at its first point does not ensure performance over its reporting range. Requiring every subsequently evaluated point to qualify changes candidates 45, 50 and 50 to 60, 70 and 70. No reversal occurs in the eight KUL3/reference seeds; no pooled candidate qualifies. These are descriptive selections on the recorded benchmark.

2.6 Patient influence and the cohort gap

Enumerating all 45 and 15 two-patient holdout pairs separates patient omission from repeated simulation. Keying streams to the sorted pair identifiers keeps every retained pair fixed. Raising the inner bootstrap budget from 200 to 1,000 and 5,000 leaves the first cohort’s candidate at 70, but moves the second’s 400 to 300 and back to 400. Omitting one source patient changes the joint coverage-and-discrimination candidate for none of ten patients in the first cohort and three of six in the second (KUL01, KUL30 and KUL31). For the coverage-only candidate, the corresponding counts are five of ten and zero of six. These checks quantify influence within each observed cohort. Appendix B gives the biological comparability limits.

The sensitivity windows cover counts 40, 50, 60, 70, 80, 90, 100, 110 in SMC and 200, 300, 400, 600, 800 in KUL3. These per-count summaries use a fixed pair schedule, so they are a separate experiment from the eight-seed union grid. Omitting KUL01 or KUL30 moves the joint candidate from 400 to 300; omitting KUL31 moves it to the lowest tested count, 200, with its lower boundary unobserved. The coverage-only baselines likewise occur at the lowest tested counts, 40 and 200. Their stability cannot locate a lower crossing outside those windows. Overlapping pairs do not add independent patients.

Expression dispersion partly explains the cohort gap. Holding tissue eligibility fixed at reference-only, the complete SMC and KUL3 mature-cell pools contain 662 and 844 cells. Across their 45 and 15 eligible patient pairs, median source-pool squared coefficients of variation ($c_P^2 = \sigma_P^2 / \mu_P^2$) are 4.35 and 16.99; median zero-expression fractions are 39.3% and 62.1%. The source means are 19.5 and 11.9 raw counts, respectively (medians over pools). The KUL3 contrast therefore has substantially more relative sampling noise despite its larger source-cell pool.

For the implemented binomial thinning, the tumour-cell variance is $s^2 \sigma_P^2 + s(1-s)\mu_P$. At $s = 0.5$, the exact conditional variance of the contrast relative to its squared signal is

$$\frac{\text{Var}(\hat{\theta} \mid P)}{\theta_P(0.5)^2} = \frac{c_P^2 + 1/\mu_P}{n_T} + \frac{4c_P^2}{n_N}. \quad (5)$$

With $n_T = 50$ and $n_N = 800$, the median signal-to-standard-deviation ratios are 3.02 in SMC and 1.53 in KUL3. This provides a distributional explanation for weaker KUL3 discrimination. It does not reproduce the exact 70-versus-400 candidate ratio: percentile-bootstrap behavior, pair heterogeneity and binning also affect that selection. The biological causes of the source-distribution difference remain unexplained.

2.7 Biological controls fail to establish mechanism

The pre-committed panel included compositional, intrinsic and retained-gene controls. In the primary cohort the retained-gene control changed by -1.000 to -0.951 relative to baseline, almost matching the compositional controls' -1.000 to -0.952 . The intrinsic control had only 0.04 CP10K, compared with 19.4 and 24.9 for the abundant compositional controls (counts per 10,000), making it uninformative for the intended test. A second cohort reproduced the pattern. Thus the application has two distinct unresolved problems: its interval rule lacks the recorded calibration support, and its control panel does not distinguish the proposed biological mechanisms. Improving one would not establish the other.

3 What a replacement validation would need to establish

The paired analysis shows why specifying the target changes an assessment: the same intervals can have different inclusion rates against a sampled reference and the generating-pool parameter. A clinical simulator likewise needs to distinguish a requested population treatment effect, a latent finite-cohort effect and an observed-data contrast. Coverage must use the target for which the interval was designed.

The earlier recovery curve admits a simpler explanation. The oracle estimate applies Equation 1 to the sampled fractions and means, exactly reproducing their empirical decomposition. This arithmetic identity explains the comparison without evaluating its uncertainty. For example, the same sample mean has about 95% or 36% coverage when only its interval width changes (Appendix C). Established trial-estimator relationships and learned-generator checks are collected there as supporting examples.

Assess the whole decision rule using the interval that would actually be reported. Record bias and error against the chosen target, null rejection, detection at the specified effect, and abstention frequency. Report both conditional interval performance among returned answers and how often a valid answer is returned. A successful effect-detection rate can accompany excessive null rejection, so the former alone cannot validate an interval. The null target here is 5% because the intervals are nominally 95%.

Separate selecting a rule from assessing it. If a scalar threshold is intended to guarantee performance above that threshold, qualification must address the evaluated upper range rather than only its first passing point. Reserve independent simulation draws for the final assessment, report rate uncertainty and the tested range, and distinguish this from transfer to new patients or cohorts. Increasing resamples from an empirical pool improves conditional numerical precision; it does not broaden that pool's biological support. For an external-control application, the additional question is comparability across data sources, which requires direct study-level validation such as the held-out comparisons of Ventz et al. [2019]. Evaluating transfer to patient populations requires that additional evidence.

3.1 A synthetic external-control example: validating the wrong population

A separate constructed experiment makes target-population mismatch consequential. A fixed synthetic registry supplies a single-arm trial and three external controls per enrolled patient. Enrolment favors heavily pretreated strata; induced benefits vary by stratum and latent individual responsiveness. The intended target is the mean individual treatment effect among enrolled records. We compare stratum contrasts standardized to the enrolled population with the same contrasts averaged over the pooled trial-plus-control file. The latter estimates a different population effect; it is defective only relative to the stated enrolled-population target. Appendix D specifies the design and controls.

At 1,600 enrolled patients, pooled standardization has a median estimate of 1.890 outcome units while the enrolled records' median true effect is 1.003. These medians fall on opposite sides of the prespecified illustrative threshold of 1.5; this is not an estimated probability of a wrong decision. Against the observed pooled contrast, the pooled estimator has zero error by construction, while enrolled standardization appears inaccurate (Table 3). Using the enrolled potential-outcome reference reverses that ranking. Merely using potential outcomes is insufficient: averaging them over the pooled population still favors the estimator of the wrong target. The target must specify both the effect and the population over which it is averaged.

Table 3: Reference choice reverses the estimator ranking in the synthetic external-control example. Bias and RMSE use 200 studies with 1,600 enrolled records and 4,800 controls. Columns identify the population used to average observed stratum contrasts; only enrolled standardization matches the intended target. All entries are in abstract outcome units.

Validation reference	Pooled standardization		Enrolled standardization	
	Bias	RMSE	Bias	RMSE
Observed contrast, pooled	0.000	0.000	-0.872	0.876
Potential outcomes, pooled	+0.003	0.115	-0.869	0.875
Potential outcomes, enrolled	+0.885	0.892	+0.013	0.105

Two controls limit the interpretation. At zero case-mix shift, pooled standardization still reproduces its observed-data reference but its bias against the enrolled effect is only -0.002 units. Equality therefore does not establish that an estimate is wrong. Under the full shift, pooled weighting balances the measured strata between arms to numerical precision while their weighted distribution differs from the enrolled distribution (maximum standardized difference 0.964). Between-arm balance does not establish target-population alignment. This constructed, uncensored outcome experiment demonstrates a possible validation failure; it supplies no evidence of its prevalence or of performance in an actual clinical trial.

4 Conclusion

The audited 50-cell rule fails joint interval evaluation despite reassuring point-estimate recovery. A larger simulation budget distinguishes three coverage failures from one passing condition, while excess null rejection and insufficient detection occur in all four. Welch intervals improve coverage but do not repair the reporting boundary. The non-monotone detection curve also becomes interpretable only beside the changing false-positive rate.

A reporting threshold must be assessed with its actual interval, explicit target and null and alternative operating characteristics. The synthetic external-control example adds that the target must name the population over which effects are averaged. These results establish failures and delimit one attempted repair; they do not validate a replacement rule or a patient world model.

5 Responsible use, limitations and impact

Misuse and limitations. NOT ESTIMABLE must not be read as “no effect” or as evidence that surviving cells are behaving normally. A separate within-cell analysis over three cohorts and 42 patients also remained unresolved because its comparability checks returned *undecided* (Appendix B). Calibration here covers one gene, one labelling axis and one weighting on two small cohorts. It challenges this rule’s justification, not cell-count rules in general. The fixed-pool analysis estimates coverage conditional on empirical cell distributions; uncertainty over new patients remains unmeasured. The draw-pool choice was underspecified, repeated simulations do not add independent patients, and the leave-one-patient-out sensitivity remains conditional on these cohorts; patient-level population uncertainty in the candidate cutpoints is unquantified. The absence-defined labelling axis remains sensitive to sequencing depth (Appendix B). The external search found no affirmative case, so the documented failure remains a self-audit rather than evidence about its prevalence. The external-control example uses a synthetic registry and an illustrative decision threshold; it does not establish clinical benefit or support a treatment decision.

Data and impact. The single-cell cohorts are public, de-identified and previously published. The release policy covers derived summaries and code, rather than cell-level matrices. A system reporting group-specific quantities should distinguish an unavailable comparison from an estimated null and should evaluate the coverage and discrimination of any abstention rule. Publishing failed calibration attempts helps prevent unsupported thresholds from being treated as validated.

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A Simulation settings

Experiment	Fixed settings and sampling units
Single-cell calibration	GSE132465 (SMC): 10 paired patients; GSE144735 (KUL3): 6. Target: GUCA2A; <code>stem_pole</code> axis, <code>lineage</code> resolution, <code>differentiated</code> mature label; reference (<code>normal</code>) weighting. Each replicate holds out two patients and draws 2,000 cells per synthetic arm from their eligible pool; reference mature fraction 0.40. Expression shifts are 1.0, 0.8, 0.5, 0.25; calibration uses 0.5.
Interval construction	A nominal 95% percentile interval from 200 resamples of mature cells within each synthetic arm, holding cell fractions fixed. This assesses the conditional intrinsic contrast; it does not estimate population uncertainty across patients. Coverage and discrimination condition on defined intervals. The historical inclusion score uses the drawn reference in Equation 2; the paired follow-up also evaluates fixed-pool coverage using Equation 3.

Historical grid budgets are reported in Appendix B; Section 2 specifies the controlled grid; the larger interval comparison is detailed next. The source cells are observed data; the external-control outcomes are simulated.

Precision follow-up and comparator. Five fresh seed streams (202609160–202609164), each with 1,000 studies per setting, provide 5,000 draws at each of four counts and two shifts in four cohort/pool combinations. Every draw uniformly chooses a two-patient pair, samples 800 reference and n_T tumour mature-cell counts independently with replacement, and thins tumour counts binomially. This is the original generator’s marginal law for this contrast, not a replay of its random stream. All 120 source-pair summaries match the earlier analysis. The vectorized percentile routine agrees with the original implementation on 32 actual-pool checks; the Welch implementation agrees with an independent library routine.

Writing $v_a = S_a^2/n_a$ for arm a , the Welch interval is $\hat{\theta} \pm t_{0.975, \nu} f_N \sqrt{v_N + v_T}$, where $\nu = (v_N + v_T)^2 / [v_N^2/(n_N - 1) + v_T^2/(n_T - 1)]$. Zero estimated total variance would give an unavailable comparator interval; none occurred. For the sparsity diagnostic, sample-based standard error is compared with $f_N \sqrt{\sigma_P^2/n_N + (s^2\sigma_P^2 + s(1-s)\mu_P)/n_T}$. Aggregate outputs retain signed rejections, widths, seed-specific rates and paired differences. The settings and single comparator were fixed before this follow-up was run; it remains an exploratory study on previously examined source data. The earlier 200-draw target diagnostic is retained in the supporting summaries.

B Calibration details and limits

Historical grids. Before the controlled run, the SMC reference-only sweeps returned binned joint candidates of 100, 90 and 70 on the committed, filled and dense grids, with 500, 500 and 200 replicates per setting, respectively. Their adjacent crossing brackets were (40, 100], (65, 90] and (42.5, 70], describing evaluated bin summaries rather than confidence intervals. The 100-to-90 change occurred within the same bin, (76, 107]: adding counts 80 and 90 to 100 changed both its median count and its aggregated outcomes. Random streams and bin boundaries also differed, so this comparison alone did not isolate a resolution effect. The controlled analysis in Section 2 supersedes it for that question. Under pooled draws no joint cutpoint qualified in either cohort’s historical grids. KUL3 reference-only candidates were 400 or 800; its largest setting sampled 800 mature cells per arm with replacement from 844 distinct mature source cells. Sampling with replacement is a valid empirical-pool simulation, but an 800-draw arm is not 800 distinct cells or additional independent patients.

Recovery accounting and unavailable contrasts. The four diagnostic sweeps comprise 1,711,200 replicates: 1,283,400 with nonzero requested intrinsic reference and 427,800 with zero requested reference, for which ratios are undefined. The first 50-replicate sweep has 119,600 draws, partitioned into 89,700 and 29,900, respectively. The maximum estimate–realised-reference residual is zero throughout. At a zero-cell setting the raw ratio is $1/(1-s)$ when the denominator is nonzero. These raw diagnostic values arise from the harness setting an empty-arm mean to zero; they do not define an effect in absent cells. The reporting gate abstains and the interval is undefined. Separately, undefined

intervals were previously counted as failures through comparisons with NaN; corrected rates exclude them and report abstentions. An all-abstaining bin has no coverage estimate. One historical seed gives $ok = wide = 45$, showing that a three-state specification need not yield three distinct fitted regions.

Biological scope and patient uncertainty. The finest annotation resolution yields abstention in 28/28 patients under one population definition and 24/28 under the other; coarser definitions do not resolve identifiability. The coarsest definition labels every epithelial cell mature and forces the compositional term to zero. In a separate comparability analysis over three cohorts and 42 patients, 83 of 84 housekeeping-detection control rows pass a 0.10 tolerance despite expression changes of 1.19–1.64 fold. Patient-bootstrap $\log_2$ expression intervals cross the accepted $[-0.5, 0.5]$ boundaries in every cohort, leaving the comparisons undecided. Depth matching discards roughly three quarters of the cells and pushes about half the remaining estimable rows below the count threshold. Within SMC, changing tissue eligibility changes the mature-cell pool from 662 to 3,824 cells. The across-cohort comparison in Section 2.6 instead holds tissue eligibility fixed. These observations restrict biological interpretation. Repeated cell draws and overlapping patient pairs do not supply independent patient replication.

C Supporting estimator and trial checks

Reference dependence. With nonempty stratum arms, saturated G-computation [Robins, 1986], IPW using empirical stratum propensities, non-cross-fitted saturated AIPW and matching using every control all reproduce $T(D) = \sum_g (n_g/n)(\bar{Y}_{1g} - \bar{Y}_{0g})$. The IPW weights recover each stratum mean; the AIPW augmentation sums to zero within stratum. OLS with additive stratum indicators instead weights contrasts by normalized $n_g \hat{p}_g(1 - \hat{p}_g)$ [Angrist, 1998, Frisch and Waugh, 1933]. Under confounded assignment at 5,000 patients, switching between these two references changes the G-computation/OLS residuals from 0/0.0663 to 0.0663/ 3.5×10^{-14} . Complete 1:1 blocks make the weights coincide. These are implementation checks of established identities. Under effect heterogeneity, OLS recovery rises from 1.003 to 1.132 while its own-reference residual stays at numerical precision.

Interval control. For 200 independent $\mathcal{N}(\theta, 1)$ observations and 2,000 draws per effect, the full-sample mean has 94.60% and 95.40% coverage at $\theta = 0, 0.5$. Quartering the interval half-width leaves the estimate and its zero residual against the empirical mean unchanged, but reduces coverage to 35.95% and 36.30%. The control illustrates why residuals cannot certify intervals.

D Synthetic external-control design

The fixed registry contains 50,000 simulated records with stratum shares (0.55, 0.30, 0.15), baseline outcomes $Y_0 \mid g \sim \mathcal{N}(\mu_g, 3^2)$, $\mu = (12, 8, 4)$, and effects $Y_1 - Y_0 = \delta_g r$, $\delta = (3, 1.5, 0.25)$, $r \sim \text{Gamma}(4, 0.25)$ (shape, scale). Responsiveness is fixed per record and hidden from estimators. Enrolment weights favor shares (0.15, 0.25, 0.60); both arms sample without replacement and are disjoint. The pooled target is a mixture of enrolled and control records, not the registry population itself. Normal outcomes can be negative: this is an abstract continuous-outcome experiment, not a fitted progression-free-survival model.

There are 200 study draws at each of $n = 50, 100, 200, 400, 800, 1600$, using one seed (20260915) and one fixed registry: 1,200 studies, eight estimators and five references yield 48,000 comparisons: 47,969 finite values and 31 exclusions from one study with an empty stratum arm at $n = 50$. The registry uses the same seed. The intended finite-cohort truth averages both potential outcomes over enrolled records; it is available to the simulator, not to estimators observing one outcome per record. Five case-mix shifts and 50 balance draws at $n = 1600$ provide controls. Threshold crossings in the saved shift table compare medians, not replicate-level decisions.

For enrolled standardization, RMSE against enrolled truth falls from 0.608 at $n = 50$ to 0.105 at $n = 1600$; pooled standardization falls from 1.082 to 0.892. The nine prespecified falsification checks did not fire, but this shows consistency with those checks, not clinical realism. The example assumes the same target-population mistake in estimator and reference; its identical recovery curve

466 may itself alert a careful analyst. It neither estimates a failure frequency nor requires a reference to
467 disagree numerically with a correct estimator on every dataset.

468 **E External audit and research context**

469 **Search and classification.** A pre-registered queue contained 41 entries; 19 were screened out and
470 seven classified within the 26 entries examined in order. An amendment before code inspection
471 relaxed unsuccessful queries; the classification rule was unchanged. Generating-parameter references
472 or quantities computed outside the estimator’s observed records give NO. Same-record references
473 require examination of inputs, weights, conditioning, nuisance fitting and observation process; YES
474 requires affirmative evidence of functional identity. All seven recorded verdicts are NO: `caus1`, `target-`
475 `trial-emulation`, `ecasim`, `PlasmodeSimulation`, `StatsPAI`, `diff-diff`, and `Auto-Empirical-Research-`
476 `Skills`. The first six use generating parameters or outside-sample quantities; the last compares
477 cross-fitted partialling-out with full-sample OLS. Its 4.0×10^{-14} residual on noiseless linear data
478 becomes 0.125 after adding outcome noise. The functions agree on the restricted noiseless support but
479 not on general noisy datasets; the NO verdict concerns shared functional form. Classification used one
480 reader, only this last repository was executed, and the sample is nonrandom. The virtual-control-arm
481 query located no examinable code. These limits preclude a prevalence estimate.

482 **Related scope.** Cell-count filters are established in pseudobulk analysis [Crowell et al., 2020];
483 `scCODA`, `Milo`, `propeller` and `DA-seq` address composition [Büttner et al., 2021, Dann et al., 2022,
484 Phipson et al., 2022, Zhao et al., 2021], while `cocoa` separates composition and expression changes
485 [Petukhov et al., 2022]. Selective prediction motivates evaluating both answers and abstentions
486 [Chow, 1970, El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017]. Conformal prediction
487 coverage [Vovk et al., 2005, Angelopoulos and Bates, 2023] does not define a missing conditional
488 population or automatically provide intervals for this intrinsic parameter.

489 **Documentary source of the case.** The anonymous review record includes the original validation
490 statement (Artifact A), interval criteria (Artifact B), and numerical context. Exact source mapping
491 and checksums remain outside the review bundle. This records one internal validation error; the
492 external search found no positive case.

493 **Implementation and provenance.** Corrections separated collapsed population definitions, aligned
494 labelling, tracked producing scripts, and repaired undefined-interval denominators. Saved tables
495 retain commit hashes and seeds; these establish traceability, not statistical validity.

496 **Use of large language models.** Language-model assistants supported coding, figures, drafting and
497 model audits under author direction. They helped identify defects and introduced some of those
498 above. Authors remain responsible for all claims. Automated checks re-derive selected results and
499 complement review; they do not verify every manuscript statement.